

8 Superposition

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Interference needs coherent sources with a stable phase difference. Constructive interference arises at path differences $n\lambda$ and destructive at $(n + \frac{1}{2})\lambda$ for in-phase sources. Diffraction spreads waves after an aperture, especially where width is comparable with wavelength. Stationary waves form nodes and antinodes from counter-propagating waves.

Worked example: follow the reasoning

In Young's double-slit experiment, $\lambda = ax/D$. If $x = 2.0$ mm, $a = 0.50$ mm, $D = 2.0$ m, then $\lambda = 5.0 \times 10^{-7}$ m.

Quick check · multiple choice

A For coherent in-phase waves, which path difference yields a bright fringe? [1]

- A $\lambda/2$
- B $3\lambda/2$
- C $\lambda/4$
- D λ

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Two slits are 0.30 mm apart and 1.5 m from a screen. Adjacent bright fringes are 3.0 mm apart. Find wavelength. [3]

Final answer: _____

Answers and explanation

Quick check: D. λ

Question 1 [3 marks]

Use $x = \lambda D/a$ [1], so $\lambda = xa/D$ [1] = $(3.0 \times 10^{-3})(3.0 \times 10^{-4})/1.5 = 6.0 \times 10^{-7} \text{ m}$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.